

The called strike zone shrank twice, and swung with its rules unchanged

Introduction

In 2025 the umpire grading buffer went from two inches outside the edge to three-quarters of an inch either side (Andrews, FanGraphs, 5 May 2025), and in 2026 regular-season ABS challenges began. No published estimate puts intervals on both steps against old-rule seasons. The zone lost width mostly in 2025 and height mostly in 2026.

Methods

I re-projected 2022-2025 pitches to Statcast's 2026 plate midpoint. Sample: 1,830,231 called pitches in 12,060 regular-season games; I fitted called-strike probability near the edge. Each season's zone is its 50 percent contour. With ABS-measured heights, the zone geometry matched 10,164 of 10,168 challenge verdicts. I measured both steps net of the 2022-2024 trend, with umpire effects partially pooled. The placebo pair, 2023 and 2024, had no change to the zone or umpire grading, although the pitch clock and shift limits arrived in 2023 and other rules changed in 2024.

Results

In pre-registered checks, the placebo failed: the zone grew 10.8 (90% CI 7.7 to 13.9) square inches, beyond its ± 3 margin, so every step is descriptive. A binned cross-check found both steps 6.0 and 7.6 square inches larger, beyond its ± 3 tolerance, and top-edge intervals undercovered in the recovery test. Every interval reflects pitch-level sampling, not season-to-season variation. From 2022 to 2023, not pre-registered, the zone shrank 10.3 (90% CI 7.3 to 13.4) square inches, so the 2025 step depends on its baseline (Table 1 footnote). Figure 1 shows the 2025 step, 23.4 (95% CI 18.6 to 28.3) square inches. The 2026 step, 33.1 (95% CI 28.7 to 37.5) square inches, exceeded every All-Star-break swing (largest 6.8 square inches). In 2025 each side came in 0.41 (95% CI 0.33 to 0.49) inches, about three times the sides' 2023-to-2024 drift (Table 1, not pre-registered). In 2026 the bottom rose 0.65 and the top fell 0.63 (95% CI 0.50 to 0.76), after rising 0.47 from 2023 to 2024 (Table 1, not pre-registered). With 2025 at the plate front, as in one published 8 to 22 (Clemens, FanGraphs, 28 April 2026), the 2025-to-2026 loss is 20.0 (95% CI 13.0 to 28.2) square inches. Both at mid-plate: 32.8 (95% CI 29.3 to 36.6) square inches.

Conclusion

The 2025 zone was already smaller than any 2022-2024 zone, before regular-season challenges began, so framing and pitch-location comparisons that start from 2025 miss that step. Under the fitted model the 2026 step was the larger, 59 percent of the two combined (95% CI 53 to 65). Those intervals omit season-to-season variation, and the gap is about one season's drift, so the difference is not established. Main limit: the failed placebo; the design cannot assign either step to its rule. The sealed-set definition and acceptance criteria were committed publicly before the sealed set was opened. That set, games from 22 September 2026 on, tests calibration once the postseason ends.

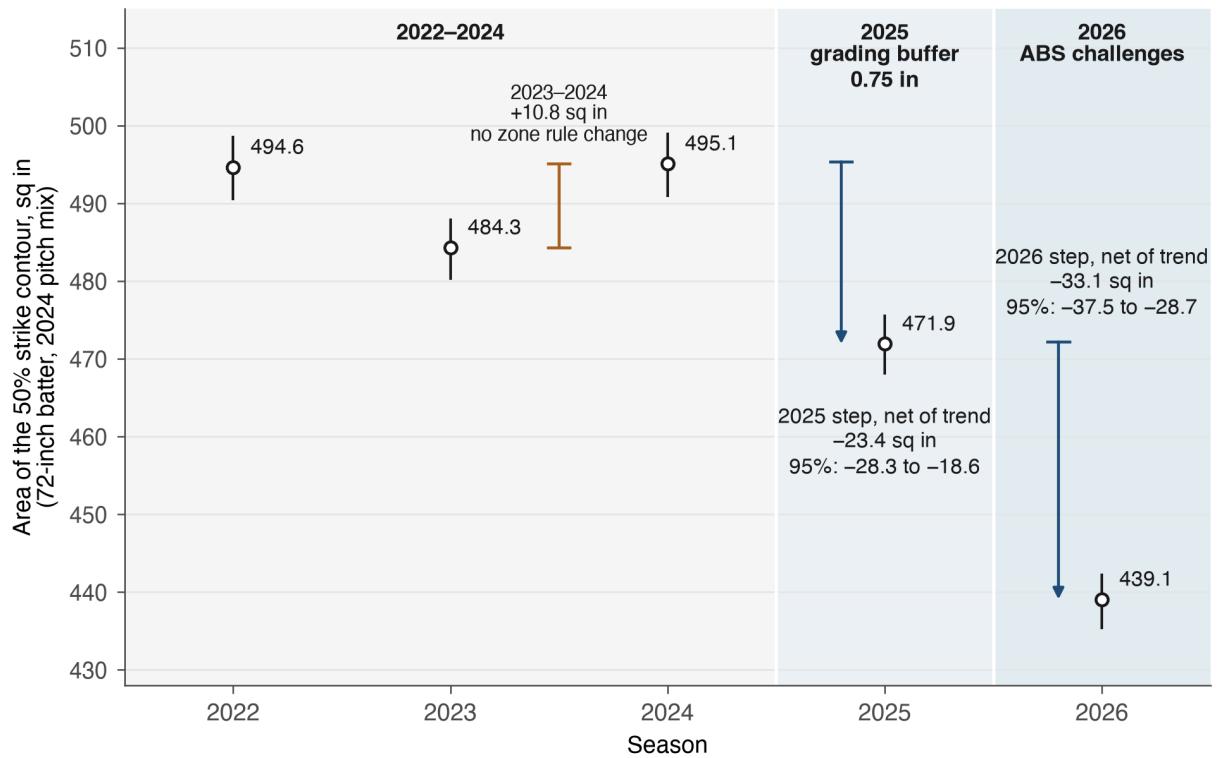


Figure 1. Points are fitted areas of the 50 percent strike contour (72-inch batter, 2024 pitch mix), with 95% intervals. Blue arrows are the 2025 and 2026 steps net of the 2022–2024 trend of +0.3 sq in per season (95%: –1.6 to 2.2). Each starts at the prior season plus the trend, so arrows and point-to-point drops differ by it. The rust bracket is the 2023-to-2024 change, 10.8 sq in (90%: 7.7 to 13.9), with zone and grading rules unchanged. That placebo failed, so period labels name no cause.

Change in the fitted 50 percent contour	Top edge (in)	Bottom edge (in)	Half-width (in)	Area (sq in)	Shadow-band strike rate (pp)
2022-2024 trend, per season	0.13 [0.09, 0.18]	0.06 [0.02, 0.09]	-0.03 [-0.06, -0.01]	0.3 [-1.6, 2.2]	-0.09 [-0.41, 0.24]
2023 to 2024, zone rules unchanged (placebo, pre-registered on area and rate; margins ± 3 sq in and ± 0.5 pp): FAILED	0.47 [0.37, 0.57] 95% interval	0.16 [0.08, 0.24] 95% interval	0.12 [0.06, 0.18] 95% interval	10.8 [7.7, 13.9] 90% interval	1.54 [1.03, 2.04] 90% interval
2025 step, net of trend (grading buffer 0.75 in)	-0.18 [-0.32, -0.05]	0.18 [0.08, 0.29]	-0.41 [-0.49, -0.33]	-23.4 [-28.3, -18.6]	-4.46 [-5.28, -3.63]
2026 step, net of trend (ABS challenges)	-0.63 [-0.76, -0.50]	0.65 [0.55, 0.74]	-0.25 [-0.32, -0.16]	-33.1 [-37.5, -28.7]	-6.05 [-6.81, -5.31]
2024 to 2026, total	-0.54 [-0.65, -0.43]	0.94 [0.85, 1.03]	-0.72 [-0.80, -0.64]	-56.0 [-59.6, -52.0]	-10.69 [-11.28, -10.04]
Measurement-plane adjustment, not a period change: 2025 read at mid-plate minus at the front	-0.77 [-0.94, -0.61]	-1.11 [-1.24, -0.98]	0.12 [-0.01, 0.25]	12.9 [6.3, 19.2]	—

Cells are point [95% interval] from coefficient draws of the fitted model; the placebo row prints its pre-registered 90% interval on area and rate, and its edge cells, added after the tag, are not pre-registered. A positive value moves an edge up or outward, or enlarges the area. pp is percentage points; the shadow band is the pitches within 3 inches, either side, of where the ball just touches the zone edge. Top-edge intervals uncovered in the recovery test. The placebo pair moved beyond its margin, so no step is attributed to its rule; the 2025 and 2026 labels name the period only. Dashes: not estimated for that row.

Exploratory, not pre-registered: against the 2022-2024 mean plus one season of trend, the 2025 area step is -19.7 [-23.3, -16.1] sq in; against 2023 alone, -12.4 [-15.8, -8.5] sq in.

Table 1. Changes in the fitted 50 percent strike contour (72-inch batter, 2024 pitch mix, mid-plate), by period: the trend, the failed placebo pair, the two steps net of trend, their total, and the measurement-plane adjustment. The note under the table defines the cells.